Arab Academy for Science, Technology, & Maritime Transport

College of Computing and Information Technology Cairo

Computer Science

B. Sc. Final Year Project

Assistify

(Intelligent Help Disk)

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DEC-25

DECLARATION

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project and to achieve those great results.

Abstract

The Assistify RAG Voice Engine project introduces intelligent customer service empowered by the latest voice recognition, large language model (LLM) technology, and Retrieval-Augmented Generation (RAG).

The system provides a means for users to communicate through both speech and text, consolidates FBI authentication techniques with role-based access control profiles, and flexes an advanced natural language understanding pipeline for robust and relevant responses from the dynamically changing organizational knowledge base.

Key functional pieces consists of FastAPI enabled backend services, the Vosk speech recognition engine, transparent LLM integration for advanced query processing, and analytic layer to capture usage and system health.

The application has been implemented with a view to maintain the system without much hassle by use of secure session management, user maintenance interfaces; on the fly knowledge base update and document upload.

Ethical considerations, privacy of data and usability are central to the design and deployment of this project, as we seek to enhance customer support efficiency as well as accessibility and trust in technology mediated service solutions.

Experimental results using both simulated and real users show improvements in response accuracy, latency of query processing and user preference.

The architecture and implementation of the project is a reference point for future AI based support systems in the enterprise and education domain.

Table of Contents

1. INTRODUCTION **7**

1.1 Purpose 7

1.2 Intended Audience 8

1.3 Intended Use 8

1.4 Scope: 10

1.5 Definitions 10

2. Literature review **11**

2.1 Related Work: *11*

2.1.1 Google Assistant: *12*

2.1.2 Amazon Alexa for Business: 13

2.1.3 IBM Watson Assistant: *14*

2.1.4 Summary: 15

2.2 SWOT Analysis: *16*

2.2.1 Strengths: 16

2.2.2 Weaknesses: 16

2.2.3 Opportunities: 16

2.2.4 Threats: 16

3 System Description & Methodology **17**

3.1 Methodology: *17*

3.1.1 User Interaction: *17*

3.1.2 Preprocessing: *17*

3.1.3. Query Routing: 17

3.1.4. Retrieval-Augmented Generation (RAG) Pipeline 17

3.1.5 Postprocessing: *18*

3.1.6 Response Delivery: *18*

3.1.7 Feedback and Analytics: *18*

3.1.8 Optional Features: *18*

3.1.9 Conclusion: 18

3.2 User Requirements and System Privileges: 18

3.2.1 Users: *18*

3.2.2 System Privileges: 19

3.3Assumptions, Constraints, Limitations, and Dependencies: 19

3.3.1. Assumptions: *19*

3.3.2. Constraints: *19*

3.3.3. Limitations: *19*

3.3.4. Dependencies: *20*

3.4 Intended Technologies: 20
3.4.1. Platforms: 20
3.4.2. Tools: 20
3.4.3. Languages: 20
4.System Requirements 21
4.1 Functional Requirements: 21
4.2 Non - Functional Requirements: 22
4.3 IEEE Standards: 23
4.4 Use Case: 34
4.5 Sequence Diagram: 35
4.6 Activity Diagram 36
4.7 Class diagram 37
4.8 UI Sketch: 38
5. References 41
. References 41

List of Figures

Figure 1 Thalmic Lab Armband 5
Figure 2 Maestro Glove 6
Figure 3 NeuroPose 7
Figure 4 Use Case Diagram 30
Figure 5 Sequence Diagram 31
Figure 6 State Machine Diagram 32
Figure 7 Activity Diagram 33
Figure 8 Home Page Sketch 34
Figure 9 Notifications Tab Sketch 34
Figure 10 Functions Page Sketch 35
Figure 11 Settings Page Sketch 35
Figure 12 Sign Up Page Sketch 36
Figure 13 Sign In Page Sketch 36

List of Tables

Table 1: Comparison between different Ai Assistants 10
Table 2: IEEE Standards Checklist 21

LIST OF ACRONYMS/ABBREVIATIONS

EMG…….Electromyography

AI…….Artificial Intelligence

TBD……To Be Done

UI…….User Interface

CNN…….Convolutional Neural Network

RNN…. Recurrent Neural Networks

VR……..Virtual Reality

AR……..Augmented Reality

SQA…Software Quality Assurance

SDLC….. software development lifecycle

IEEE…….. Institute of Electrical and Electronics Engineer

Chapter I

1. INTRODUCTION

In today's rapidly evolving digital landscape, the demand for responsive, intelligent customer support solutions is at an all-time high. Organizations are seeking ways to streamline user interactions, resolve queries efficiently, and deliver accurate information in real time.

The emergence of artificial intelligence and natural language technologies has transformed the potential for automated support, enabling machines to understand and converse using both voice and text.

The Assistify RAG Voice Engine project addresses these challenges by integrating state-of-the-art voice recognition, Retrieval-Augmented Generation (RAG), and large language models into a unified customer support platform. Designed for accessibility and scalability, the system supports multiple user roles, secure authentication, and seamless interactions through intuitive web and voice interfaces.

Central to this project is the application of ethical design—prioritizing user privacy, transparency, and unbiased service. By leveraging real-time analytics, automated knowledge base management, and open-source technologies, Assistify not only enhances user experience but also reduces operational costs and increases reliability.

This document provides a comprehensive overview of the Assistify RAG system's architecture, requirements, implementation methodology, and evaluation. The goal is to deliver a practical model for AI-driven customer support that can be adapted in a wide range of organizational contexts, fostering innovation while maintaining ethical and professional standards.

This introduction situates your project in a real-world context, articulates the motivation, technological and ethical foundations, and outlines the structure of your documentation—demonstrating both technical and human relevance.

1.1 Purpose

The primary purpose of the Assistify RAG Voice Engine project is to revolutionize the customer support experience by leveraging the latest advances in artificial intelligence, natural language processing, and voice recognition.

This system is designed to provide users with fast, accurate, and context-aware assistance through both voice and text channels, reducing wait times and lowering the burden on human support agents.

1.2 Intended Audience

The Assistify RAG Voice Engine project is crafted to engage and benefit a wide array of users and stakeholders.

Foremost, it appeals to technology enthusiasts and forward-thinking innovators curious about the intersection of artificial intelligence, natural language processing, and voice-enabled human-computer interaction.

Individuals who seek more convenient, efficient, and intuitive ways to interact with digital platforms will find strong value in this project, especially as it simplifies and enhances daily customer support experiences.

A special emphasis has been placed on accessibility; users with physical disabilities or restricted mobility are empowered to access support and information more freely, advancing inclusivity in digital technology.

Academics and researchers in AI, robotics, data science, and human-computer interaction may find this system valuable for its new approach to customer communication and ethical machine interfaces.

Administrators and engineers responsible for deploying and maintaining customer support platforms will benefit from the robust design and analytics capabilities.

Ultimately, Assistify is intended for anyone interested in the evolution of user-centered, intelligent service systems—from business leaders aiming for improved operational efficiency, to everyday users eager to experience seamless and trustworthy digital assistance.

This language reflects the audience definition approach recommended for technical documentation, balancing inclusivity, specificity, and clarity

1.3 Intended Use

The Assistify RAG Voice Engine project is designed to showcase an innovative, intuitive, and user-friendly approach to interacting with digital systems—focusing on making customer support, knowledge access, and system control simpler and more natural for everyone.

While current capabilities center on voice-enabled and text-based support, the modular design makes Assistify an ideal platform for extending into more physical and gesture-based control applications. For example, future enhancements may integrate EMG armband technology, enabling users to execute computer commands and navigate interfaces using simple hand or muscle gestures.

In its present form, Assistify empowers users to:

- Obtain answers and support rapidly by speaking or typing, without navigating complex menus.
- Seamlessly upload organizational knowledge for immediate use in the support system.
- Monitor and analyze usage through an integrated admin dashboard.

The system is suited for a wide range of applications:

- Streamlining daily customer support tasks in commercial, educational, or governmental settings.
- Enhancing self-service and accessibility for users with physical disabilities.
- Serving as a demonstrator for experimental human-computer interaction methods, e.g., gestural or wearable device control, as well as AI-driven UX research.

Ultimately, Assistify aims to break down barriers between humans and machines, offering a system that adapts to users' needs—whether through voice, text, or advanced gesture recognition. By doing so, it sets a foundation for future tools that increase independence, efficiency, and enjoyment in the digital experience.

1.4 Scope:

Asstify aims to emulate the help desk experience by using AI to answer user questions faster and more accurately through both voice and text. It focuses on making support more accessible, reducing waiting time, and decreasing the workload on human agents by handling common and repeated queries.

The main objectives are to provide a natural, user-friendly interface using voice-to-text and text-to-voice, to use LLM and RAG to give context-aware answers from a knowledge base, and to support multi-platform access such as web or mobile. The system also aims to collect user feedback to keep improving response quality and adapt to new issues over time.

The goals of Asstify include increasing customer satisfaction, lowering support costs by automating routine tasks, and offering a scalable solution that can grow with the organization's needs. It also aims to support multiple languages in the future to serve a wider range of users.

1.5 Definitions

- Asstify: A voice-enabled AI help desk assistant designed to provide quick and accurate support by combining large language models (LLMs), retrieval-augmented generation (RAG), voice-to-text (V to T), and text-to-voice (T to V) technologies. It helps users interact naturally using voice or text to get answers and resolve issues.
- AI Assistant: Software powered by artificial intelligence that understands natural language commands through voice or text, processes them, and performs tasks or provides information to assist users.
- Large Language Model (LLM): A type of AI trained on vast amounts of text data to understand and generate human-like language, enabling natural communication and response generation.
- Retrieval-Augmented Generation (RAG): A hybrid AI technique that improves response accuracy by combining information retrieval from external knowledge bases with LLM-based text generation.
- Voice-to-Text (V to T): Technology that converts spoken language into written text, allowing voice commands to be processed by the system.
- Text-to-Voice (T to V): Technology that converts written text responses into spoken voice output to enable natural audio feedback to users.

Chapter II

2. Literature review

2.1 Related Work:

In the market for AI-powered help desk and virtual assistant services using large language models (LLMs), Retrieval-Augmented Generation (RAG), and voice-to-text

(V to T) and text-to-voice (T to V) technologies, there are several notable competitors:

1- Google Assistant
2- Amazon Alexa for Business
3- IBM Watson Assistant

Advantages of Asstify:

- Multimodal Interaction: Supports both voice and text inputs, making it easy for users to interact in their preferred way.
- Real-Time Voice Processing: Includes Voice-to-Text (V to T) and Text-to-Voice (T to V) modules for seamless communication and accessibility.
- User Feedback Loop: Collects user feedback to improve the system continuously through analytics and retraining.
- Scalable and Flexible: Designed to support multiple platforms like web and mobile, and can handle escalating complex queries to human agents if needed.

Disadvantages of Asstify:

- Dependence on Internet: The system requires an internet connection for voice recognition, data retrieval, and LLM processing, which may limit offline usability.
- Complexity of Integration: Combining voice-to-text, retrieval, and large language models can be complex and may require significant resources to maintain and optimize.
- Limited Accuracy for Noisy Environments: Voice recognition quality can drop in noisy or unclear audio conditions, leading to misunderstandings or errors.

Conclusion:

The competitors mentioned, including Google Assistant, Amazon Alexa, IBM Watson Assistant, and OpenAI Whisper with ChatGPT, demonstrate the importance of combining voice interaction with powerful language models and retrieval systems. These solutions show how advanced AI can improve help desk and customer support services by enabling natural communication, quick access to information, and context-aware responses. Our project builds on these concepts to provide a tailored, voice-enabled help desk assistant that integrates LLM, RAG, and voice-to-text/text-to-voice technologies to create a modern, efficient, and user-friendly experience

2.1.1 Google Assistant*:*

Google Assistant offers advanced voice recognition, conversational AI, and integration with many platforms. It supports natural language understanding and real-time responses, making it popular for customer support and help desk applications.

Figure 1 Google Assistant

Advantages:

- Google Assistant is very good at understanding natural language and voice commands, making user interactions smooth and easy.
- It works well across many devices and platforms, like smartphones, smart speakers, and computers.
- It has a large knowledge base and can answer a wide range of questions quickly.
- Integration with Google's ecosystem (Calendar, Maps, Gmail) helps provide useful personalized assistance.

Disadvantages:

- Google Assistant requires a stable internet connection for most features, so it doesn't work well offline.
- Privacy concerns are raised since it collects a lot of user data to improve its services.
- Sometimes it struggles with complex or very specific requests outside of its pre-programmed abilities.
- It may have limited support for some local languages and dialects depending on the region.

2.1.2 Amazon Alexa for Business*:*

Alexa provides voice-activated assistance tailored for enterprise environments. Amazon's platform includes skills that can automate tasks and support queries, with strong voice input/output handling.

Figure 2 Amazon Alexa

Advantages:

- Alexa for Business offers strong voice interaction capabilities tailored for workplace use, helping automate tasks and manage meetings efficiently.
- It integrates well with many third-party business applications and smart office devices for a connected environment.
- Supports multiple users and personalized experiences through user profiles.

Disadvantages:

- Alexa for Business depends heavily on cloud services, so it requires a reliable internet connection to function properly.
- Privacy and data security concerns are important since sensitive business information might be handled through voice commands.

2.1.3 IBM Watson Assistant:

IBM Watson Assistant is an enterprise-grade conversational AI platform that enables language understanding, dialogue management, and integrates with knowledge bases via RAG techniques. It supports multilingual voice input and real-time response generation.

Figure 3 IBM Watson Assistant

Advantages:

- IBM Watson Assistant offers powerful AI-driven conversational capabilities with strong natural language understanding.
- It can integrate well with existing enterprise systems and knowledge bases, making it suitable for complex business environments.
- Supports multilingual conversations, which helps serve global users effective
  .

Disadvantages:

- Setting up and customizing Watson Assistant can require significant time and technical expertise.
- It may be costly for smaller businesses due to pricing and resource requirements.
- Some users report that response times can be slower for complex queries that need deeper processing.

Table 1: Comparison between different AI Assitants

Source	Primary Focus	Voice and Text	Customization	Deployment & Integration
Assistify	Voice-enabled AI help desk assistant with LLM & RAG for customer support	Supports both voice (V to T) and text inputs & outputs (T to V)	Can be tailored for specific help desk knowledge bases and workflows	Web and mobile platforms, integrates with support databases via RAG
Google Assistant	General voice assistant for consumers and smart devices	Excellent voice recognition, with some text interface	Limited customization, mostly general use	Works across many devices (phones, smart speakers)
Amazon Alexa	Business and workplace voice assistant focused on productivity	Strong voice capabilities for business environments	Custom skills and integrations for enterprise needs	Integrated with many business tools and smart office systems
IBM Watson Assistant	Enterprise conversational AI for complex business needs	Multimodal with strong NLP and integration with text-based systems	Highly customizable, supports complex domain-specific training	Integrates deeply with enterprise software and data systems

2.1.4 Summary:

This comparison shows that Asstify is specially designed as a voice-enabled help desk assistant, focusing on providing accurate customer support using advanced AI techniques like large language models (LLM) and retrieval-augmented generation (RAG). Google Assistant is a popular general-purpose voice assistant widely used across many devices, offering great voice recognition but with less customization for specific business needs. Amazon Alexa for Business targets workplace productivity with voice automation and integration in office environments but requires technical setup for customization. IBM Watson Assistant is a powerful enterprise AI solution that can handle complex conversations and integrates deeply with business systems but demands more resources for deployment and training. Overall, Asstify stands out by combining voice and text interaction with tailored knowledge base integration for modern help desk solutions.

2.2 SWOT Analysis:

2.2.1 Strengths:

- Asstify uses advanced AI with large language models and retrieval-augmented generation, making responses accurate and context aware.
- It supports both voice and text, so users can interact in the way they feel most comfortable.
- Its feedback system allows continuous improvement based on real user interactions.

2.2.2 Weaknesses:

- Asstify depends on a good internet connection, so it might not work well in offline or weak network areas.
- The system's complexity means it needs careful maintenance and technical knowledge to keep it running smoothly.
- Voice recognition accuracy may drop in noisy environments or with unclear speech.
- Privacy concerns may arise because of handling sensitive voice and text data.

2.2.3 Opportunities:

- Growing demand for AI-powered help desk solutions can help Asstify expand into many industries.
- Integration with more platforms like mobile apps and smart devices could increase usability.
- Continuous advances in AI and voice tech will improve Asstify's performance and features

2.2.4 Threats:

- Strong competitors like Google Assistant, Alexa, and IBM Watson have more resources and wider user bases.
- Rapid AI technology changes might require quick adaptation to stay relevant.
- Privacy laws and data regulations could limit how Asstify uses voice and text data.

Chapter III

3 System Description & Methodology

3.1 Methodology:

Asstify combines advanced AI technologies to emulate help desk support by enabling users to interact via voice or text. It uses voice-to-text and text-to-voice modules for flexible communication. The core AI consists of a large language model (LLM) augmented with retrieval-augmented generation (RAG) to deliver context-aware and accurate responses by searching a knowledge base. Queries are first analyzed for intent, then routed either to direct LLM response or the RAG pipeline for more complex cases. The system supports multi-platform deployment, continuous learning through user feedback, and optional features like image processing and human escalation. This approach ensures a scalable, efficient, and user-friendly help desk experience.

3.1.1 User Interaction:

Users can submit their support requests through multiple channels such as web, mobile, or API. Inputs can be voice, text, or optionally images like screenshots for troubleshooting.

3.1.2 Preprocessing:

Voice inputs are converted to text using the Voice-to-Text (V to T) module. Basic language detection and intent recognition are applied to understand the user's request.

3.1.3. Query Routing:

Based on the user's intent, the system decides whether to answer directly via the large language model (LLM) for simple questions or send the query to the Retrieval-Augmented Generation (RAG) pipeline for detailed, context-dependent answers or escalation.

3.1.4. Retrieval-Augmented Generation (RAG) Pipeline

Data from internal documents and support tickets are prepared by chunking and embedding into vector form.

These embeddings are stored in a vector database for fast retrieval.

The user query is also embedded and used to search for relevant document chunks.

These chunks are combined with the user query to form a detailed prompt for the LLM to generate an accurate response.

3.1.5 Postprocessing:
Text responses are formatted, simplified, or translated if needed. For voice requests, text responses are converted back to speech via the Text-to-Voice (T to V) module.

3.1.6 Response Delivery:
The final response is shown on the user's interface, such as chat UI or voice playback, with options for user feedback or corrections.

3.1.7 Feedback and Analytics:
User feedback is collected to analyze satisfaction and improve system accuracy continuously through retraining and updating.

3.1.8 Optional Features:
Include image input processing with OCR, user authentication, multi-language support, and human agent escalation for unresolved or complex issues.

3.1.9 Conclusion:
Asstify uses advanced AI to closely mimic a traditional help desk, enabling users to interact naturally through voice or text. It combines language models with retrieval techniques to find precise answers from a knowledge base and smartly routes queries based on their complexity. Continuous learning from user feedback and optional features like image processing and human support escalation make Asstify an effective and flexible virtual assistant for customer service.

3.2 User Requirements and System Privileges:

3.2.1 Users:
The system is designed for a diverse user base, including technology enthusiasts, innovators, individuals with physical disabilities, gamers, researchers in the fields of robotics and artificial intelligence, and wearable technology.

3.2.1.1 User Characteristics and Roles:

- End Users: General customers seeking support, may have diverse technical skill levels; use voice or text to interact.
- Help Desk Agents: Trained support staff handling escalations and complex queries beyond AI scope.
- Administrators: Technical or managerial staff managing configuration, users, analytics, and knowledge base maintenance.

3.2.1.2. Features Available for Each User:

- End Users: Submit queries, receive AI-generated responses, provide feedback, request human assistance.
- Help Desk Agents: Access active sessions, respond to escalations, update support content, monitor issues.
- Administrators: Manage users/roles, configure system settings, oversee analytics, update backend data.

3.2.2 System Privileges:

Role	**Privileges**
End Users	Query submission, response viewing, feedback submission, escalation requests
Help Desk Agents	Session access, response handling, content/knowledge base update
Administrators	Full management rights including user roles, system config, and analytics review

3.3Assumptions, Constraints, Limitations, and Dependencies:

3.3.1. Assumptions:

- Users have internet access to interact with Asstify.
- Queries can be interpreted by language models and voice recognition systems.
- Knowledge base content is regularly updated and accurate.

3.3.2. Constraints:

- System performance depends on internet speed and server uptime.
- Voice recognition accuracy may decrease in noisy environments.
- Human agent intervention required for complex or unclear queries.

3.3.3. Limitations:

- Limited offline functionality due to reliance on cloud AI services.
- Potential privacy concerns handling sensitive user data.

3.3.4. Dependencies:

- Integration with vector databases for retrieval-augmented generation.
- Access to large language model APIs and voice processing modules.
- Compatibility with user devices and browsers.

3.4 Intended Technologies:

3.4.1. Platforms:

- Web browsers (Chrome, Edge, Firefox)
- Mobile apps on iOS and Android (if developed)

3.4.2. Tools:

- FastAPI backend framework
- Vosk or similar for voice-to-text
- Text-to-speech engines (e.g., Google TTS)
- Vector databases (Milvus, Pinecone)

3.4.3. Languages:

- Python for backend and AI processing
- JavaScript/HTML/CSS for frontend UI

3.4.4. Communication Protocols:

- HTTP/HTTPS for API calls
- WebSocket for real-time communication between client and server

Chapter IV

4.System Requirements

4.1 Functional Requirements:

- User Interaction:
  Allow users to submit queries via voice or text input on web and mobile platforms.
- Voice Processing:
  Convert voice inputs to text accurately using voice-to-text technology.
- Intent Recognition & Routing:
  Analyze queries to determine whether to respond directly or route to retrieval-based AI for detailed answers.
- Information Retrieval:
  Search knowledge bases using semantic vector search for relevant information.
- Response Generation:
  Generate context-aware, natural language responses combining retrieved data with large language models.
- Voice Output:
  Provide text-to-voice conversion to deliver spoken responses when requested.
- User Feedback Collection:
  Collect user ratings and comments to improve response accuracy and system training.
- Human Escalation:
  Escalate complex or unresolved queries to human agents seamlessly.
- Administrator Controls:
  Manage users, roles, system settings, and knowledge base updates.
- Real-time Communication:
  Support real-time interaction and updates using WebSocket or similar protocols.

4.2 Non - Functional Requirements:

Performance:

- The system must provide fast and responsive interactions, minimizing latency in voice recognition, query processing, and response delivery to ensure a smooth user experience.
- It should handle multiple concurrent users efficiently without degradation in response times.
- Voice-to-text and text-to-voice conversions need to operate in near real-time to simulate natural conversations.

Integration:

- Asstify must seamlessly integrate with existing knowledge bases, ticketing systems, and CRM platforms to retrieve and update relevant information.
- Integration with third-party services such as voice recognition and speech synthesis APIs should be reliable and secure.
- The system should support standard communication protocols (HTTP/HTTPS, WebSocket) for interoperability across diverse client platforms.

Usability:

- User interfaces must be intuitive and easy to navigate for users with varied technical backgrounds.
- Provide multi-language support and accessibility features to accommodate diverse users.
- Enable clear voice prompts and error handling to guide users effectively through interactions.

Reliability:

- Asstify must maintain high availability with minimal downtime to ensure consistent access for users.
- The system should be fault-tolerant, gracefully handling errors or failures without crashing or losing user data.
- Robust backup and disaster recovery mechanisms should be in place to protect stored knowledge bases and interaction logs.
- Monitoring and alerting systems should detect and respond quickly to any performance or security issues to maintain service continuity.

Interface:

- The GUI should be responsive on different screen sizes.

Language:

- Available only in English.

4.3 IEEE Standards:

Table 2 IEEE Standards Checklist

IEEE standards	Description	Checklist	Satisfied in
730 (Software quality assurance)	**Introduction:** ● Introduce the SQA plan. ● Outline the purpose, scope, and objectives of the plan.	X X	**TBD In Project 2**
	References: ● Include a list of all the documents referenced in the SQA plan. ● Definitions and Acronyms: ● Define any specific terms or acronyms used in the document to ensure clarity.	X X X	**TBD In Project 2** **TBD In Project 2**
	Overall Description: ● Describe the project, its objectives, and the software to be developed or maintained. ● Outline the project's organizational structure and identify key personnel.	✔ X	**Chapter 1 Section 1.1** **TBD In Project 2**
	Document Organization: ● Provide an overview of the structure of the SQA plan, outlining the major sections and their contents.	X	**TBD In Project 2**
	Responsibilities: ● Define the roles and responsibilities of individuals and groups involved in the SQA activities. ● Specify the authority and independence of the SQA team.	X X X	**TBD In Project 2** **TBD In Project 2**
	SQA Tasks:	X	**TBD In Project 2**

	● Detail the specific SQA tasks that will be performed throughout the SDLC . ● Include activities such as reviews, audits, and testing.	X	**TBD In Project 2** **TBD In Project 2**
	SQA Schedule: ● Provide a schedule for the SQA tasks, indicating when each task will be performed during the SDLC.	X	**TBD In Project 2**
	SQA Resources: ● Identify the resources required for SQA activities, including personnel, tools, and facilities.	X	**TBD In Project 2**
	Training: ● Outline plans for training personnel involved in SQA activities, ensuring they possess the necessary skills and knowledge.	X X	**TBD In Project 2**
	SQA Records: ● Define the types of records that will be generated during SQA activities and outline their retention and access policies.	X X	**TBD In Project 2** **TBD In Project 2**
	Problem Reporting and Corrective Action: ● Establish procedures for reporting and tracking software defects and issues.	X X	**TBD In Project 2** **TBD In Project 2**

	● Detail the corrective action process to address identified problems.	X	
	Tools, Techniques, and Methodologies: ● Identify the tools, techniques, and methodologies that will be used to perform SQA activities.	X	**TBD In Project 2**
	Quality Assurance Metrics: ● Define the metrics that will be used to measure the effectiveness of the SQA activities.	X	**TBD In Project 2**
	Supplier Control: ● If applicable, outline the procedures for evaluating and controlling the quality of externally provided components or services.		**TBD In Project 2**
	Documentation: ● Specify the documentation standards and procedures that will be followed during the project.		
	Non-Compliance and Corrective Action: ● Describe the actions to be taken in case of non-compliance with the SQA plan and outline the corrective measures.		

830 (software requirements)	**Introduction:** ● Provide an overview of the software system. ● Specify the purpose and scope of the document. ● Identify the intended audience (e.g., developers, testers, users).	✔ ✔ ✔	**Chapter 1** **Chapter 1 Section 1.1,1.4** **Chapter 1 Section 1.2**
	General Description: ● Describe the overall product, including its functions, features, and characteristics. ● Discuss the context within which the software will be used. ● Define any relevant terminology or acronyms.	✔ ✔ ✔ ✔	**Chapter 4 Section 4.1,4.2** **Chapter 1 Section 1.3**
	Specific Requirements: ● Detail the functional requirements, specifying the system's capabilities and functions. ● Include non-functional requirements, such as performance, reliability, usability, and security. ● Use clear and concise language to express each requirement. ● Use a consistent format for presenting requirements.	✔ ✔ ✔ X	**Chapter 4 Section 4.1** **Chapter 4 Section 4.2**
	External Interface Requirements: ● Specify how the software interacts with external entities, including users, hardware, software, and other systems. ● Detail input and output formats, data structures, and protocols.	X X	**TBD In Project 2** **TBD In Project 2** **TBD In Project 2**
	System Features: ● Provide a detailed description of each system feature.	X X	**TBD In Project 2**

	● Include information on inputs, processes, and outputs for each feature.	X	**TBD In Project 2**
	Performance Requirements: ● Define the performance criteria that the software must meet. ● Include information on response time, throughput, and other relevant performance metrics.	✔ ✔	**TBD In Project 2** **Chapter 3 Section3.3**
	Design Constraints: ● Identify any constraints that may affect the design or implementation of the software. ● This could include limitations on hardware, software, or other resources.	✔ ✔	**Chapter 4 Section 4.2**
	Quality Attributes: ● Specify any quality attributes that are important for the software, such as reliability, maintainability, and scalability.	✔	**Chapter 4**
	Other Requirements: ● Include any other requirements that do not fit into the categories mentioned above.		**Chapter 4 Section 4.4,4.5.4.6,4.7 Chapter 5**
	Appendix: ● Provide any additional supporting information, such as diagrams, charts, or reference materials.		

1012 (Software specification and verification)	**Scope and Purpose:** ● Clearly define the scope and purpose of your verification and validation processes. This should include the objectives, constraints, and any specific goals related to your project.	**X**	**TBD In Project 2**
	Life Cycle Processes: ● Describe the life cycle processes that you will follow for verification and validation. IEEE 1012 emphasizes a systematic and well-defined approach to these processes.	**X**	**TBD In Project 2**
	Activities and Tasks: ● Identify and define the specific activities and tasks involved in verification and validation. This may include requirements analysis, design evaluation, testing, and other related processes.	**X** **X** **X**	**TBD In Project 2** **TBD In Project 2** **TBD In Project 2**
	Documentation: ● Address the documentation requirements outlined in IEEE 1012. This involves creating and maintaining documents that describe the verification and validation activities, plans, and results.	**X**	**TBD In Project 2**
	Verification Planning: ● Develop a verification plan that outlines the approach, resources, schedule, and criteria for verification activities. This plan should align with the overall project plan.	**X**	**TBD In Project 2**
	Validation Planning:		

	● Similarly, create a validation plan that covers the approach, resources, schedule, and criteria for validation activities. Ensure that your validation plan is consistent with your project goals and requirements.	X	**TBD In Project 2**
	Configuration Management: ● Implement configuration management practices for your software and related artifacts. This includes version control, change management, and maintaining baselines for different stages of development.	X	**TBD In Project 2**
	Test Planning and Execution: ● Detail your test planning and execution processes. Define test cases, scenarios, and criteria for acceptance. Ensure that testing is comprehensive and covers various aspects of the software.	X	**TBD In Project 2**
	Analysis and Evaluation: ● Incorporate methods for analysing and evaluating the results of verification and validation activities. This includes identifying defects, assessing the quality of the software, and making decisions based on the evaluation.	X	**TBD In Project 2**
	Reporting: ● Define how you will report the results of verification and validation activities. Provide documentation that summarizes	X	**TBD In Project 2**

	findings, issues, and any necessary corrective actions. **Quality Assurance:** ● Implement quality assurance measures throughout the software development life cycle. This includes adherence to coding standards, documentation practices, and other relevant guidelines. **Reviews and Audits:** ● Conduct regular reviews and audits of the software and related documentation to ensure compliance with quality standards and project requirements.	**X**	**TBD In Project 2**
1016 (software design descriptions)	**Introduction:** ● Introduce the design description document. ● Identify the software for which the design is being described. ● Include any relevant background information or context. **System Overview:** ● Describe the purpose and scope of the software system. ● Provide an overview of the system architecture and its components. ● Include any relevant diagrams or charts to illustrate the system's structure. **Architectural Design:**	X **X** X ✔ X X **X**	**TBD In Project 2** **Chapter 1 Section 1.1,1.4** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2**

	● Present the overall architecture of the software system. ● Describe the key architectural components and their interactions. ● Include diagrams, such as architectural block diagrams or UML diagrams. ● Discuss design decisions and justifications. **Detail Design**: ● Provide detailed information on each component identified in the architectural design. ● Include detailed specifications for modules, classes, or components. ● Describe data structures, algorithms, and interfaces. ● Use appropriate diagrams to illustrate the detailed design. **Data Design:** ● Describe the data architecture, including databases, data structures, and data flow. ● Specify data formats, storage, and access mechanisms. ● Use diagrams such as entity-relationship diagrams or data flow diagrams. **Component-Level Design:** ● Detail the design of individual components/modules. ● Specify interfaces, dependencies, and interactions. ● Include pseudocode or code snippets to illustrate key aspects. **Human Interface Design:**	**X** **X** **X** **X** **X** X X X X X X **X** **X** ✔ ✔ ✔	**TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **Chapter 4 Section 4.8**

	● Describe the design of the user interface if applicable. ● Include details on user interactions, screen layouts, and navigation. ● Use mockups or prototypes to illustrate the design. **Dependencies:** ● Identify dependencies between components, modules, and external systems. ● Discuss how changes in one part of the system may impact other parts. **Design Rationale:** ● Provide justifications for design decisions. ● Explain trade-offs and alternatives considered during the design process. **Verification and Validation:** ● Outline the strategies for verifying and validating the design. ● Discuss any testing or simulation approaches used to ensure the design's correctness. **Interfaces:** ● Document the external interfaces of the system, including APIs, libraries, and communication protocols. ● Specify the expected inputs and outputs for each interface. **Constraints:** ● Identify any constraints that influenced the design process, such as hardware limitations,	X X X X X X X X	**TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2**

	regulatory requirements, or performance constraints.		
1219 (Software Maintenance)	**Introduction :** ● Overview of the software maintenance processes. **Maintenance Processes** ● Descriptions of the processes used for maintaining the software after it is released. **Maintenance Documentation :** ● Plans for creating and maintaining documentation related to software maintenance. **Problem and Modification Analysis :** ● Processes for analyzing problems and implementing modifications. **Tools, Techniques, and Methods:** ● Tools and techniques used in the maintenance process.	**X** **X** **X** **X** **X**	**TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2** **TBD In Project 2**

4.4 Use Case:

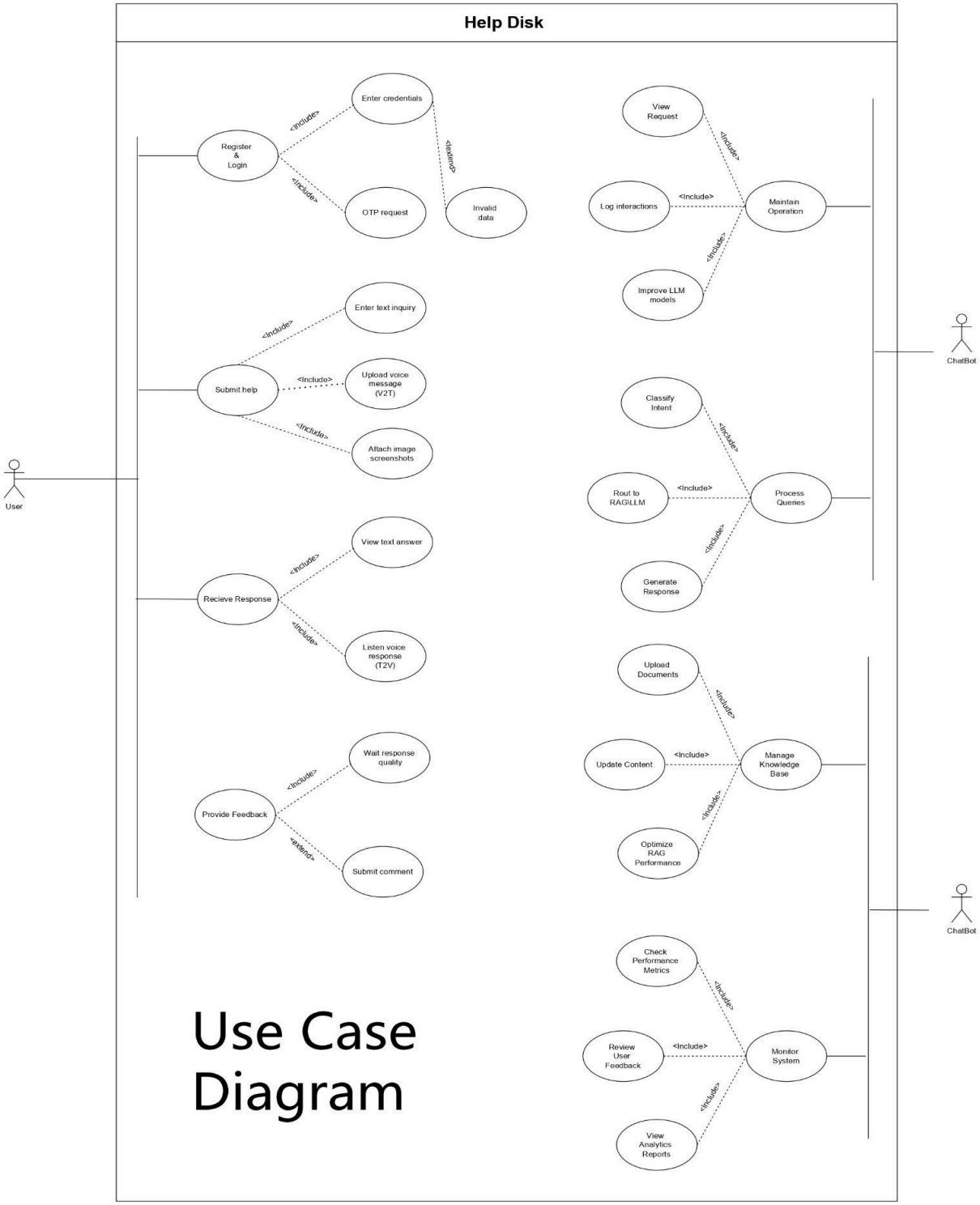

Figure 4 Use Case Diagram

4.5 Sequence Diagram:

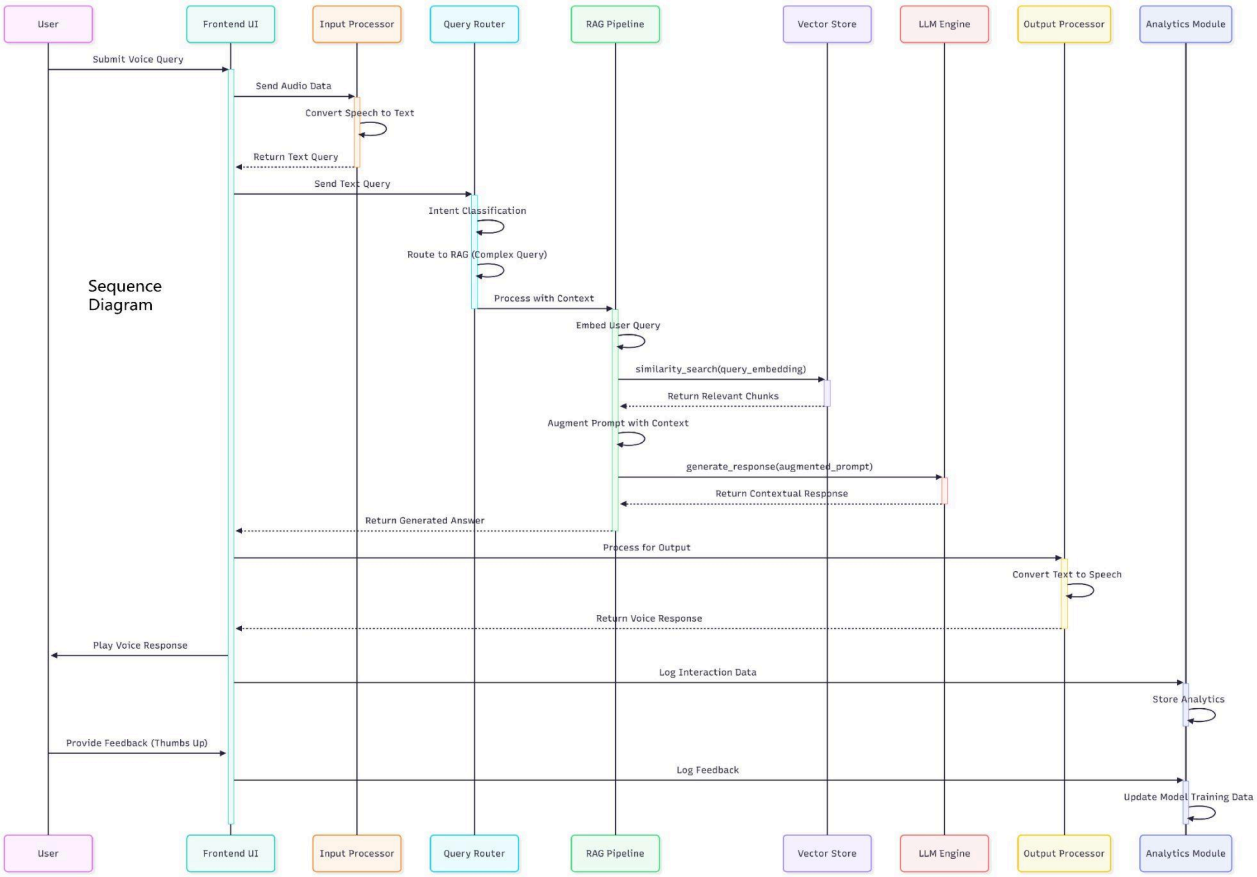

Figure 5 Sequence Diagram

4.6 Activity Diagram

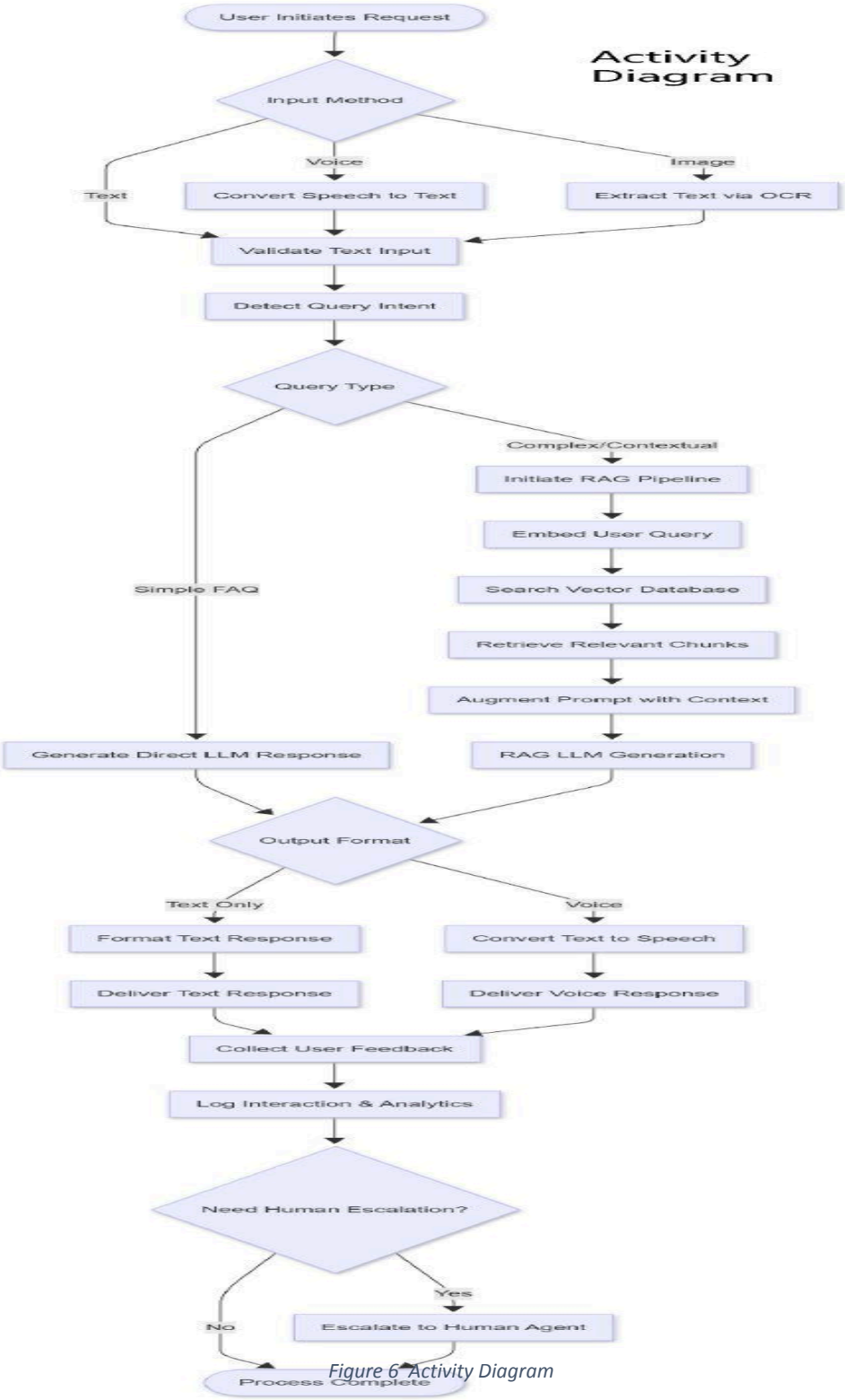

Figure 6 Activity Diagram

4.7 Class diagram

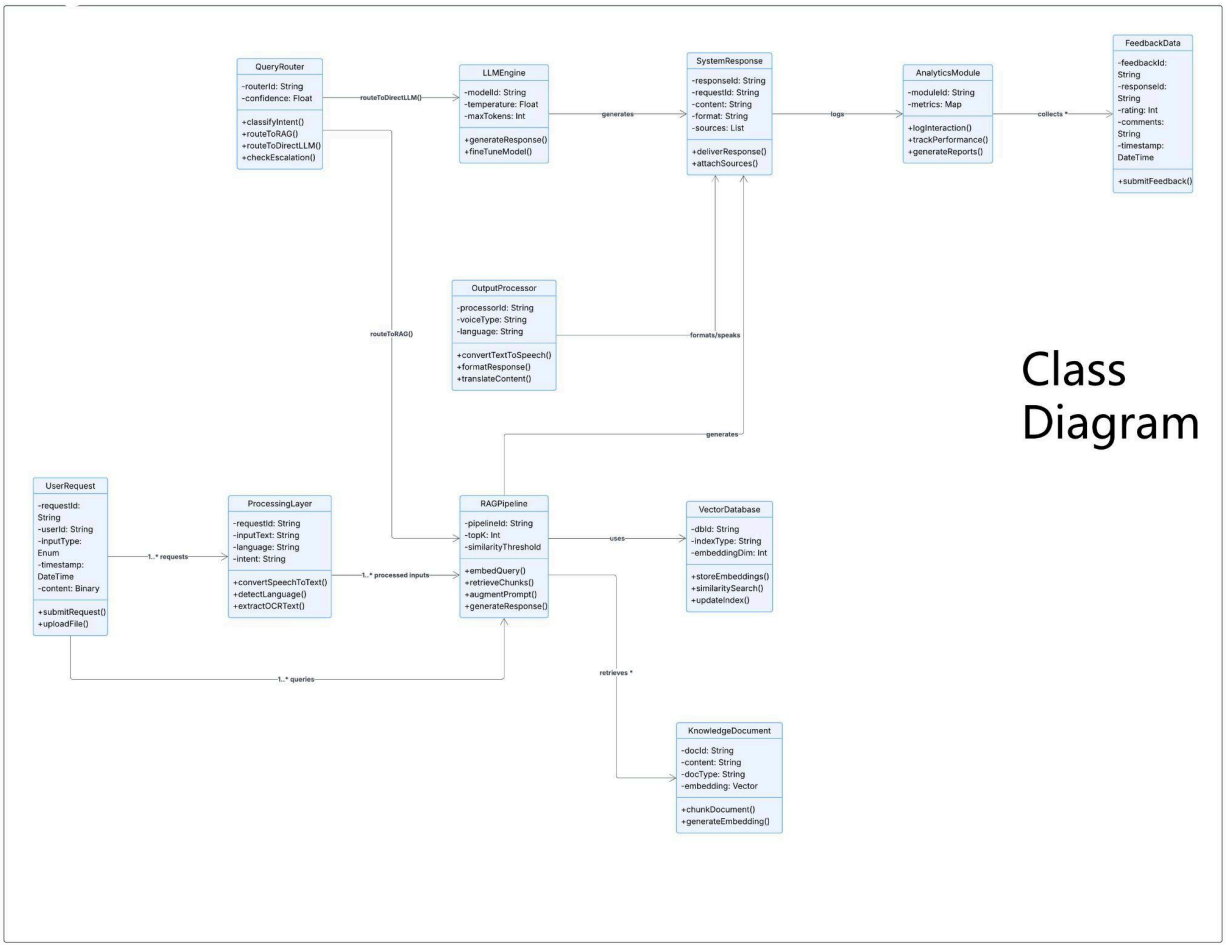

Figure 7 Class Diagram

4.8 UI Sketch:

Home Page:

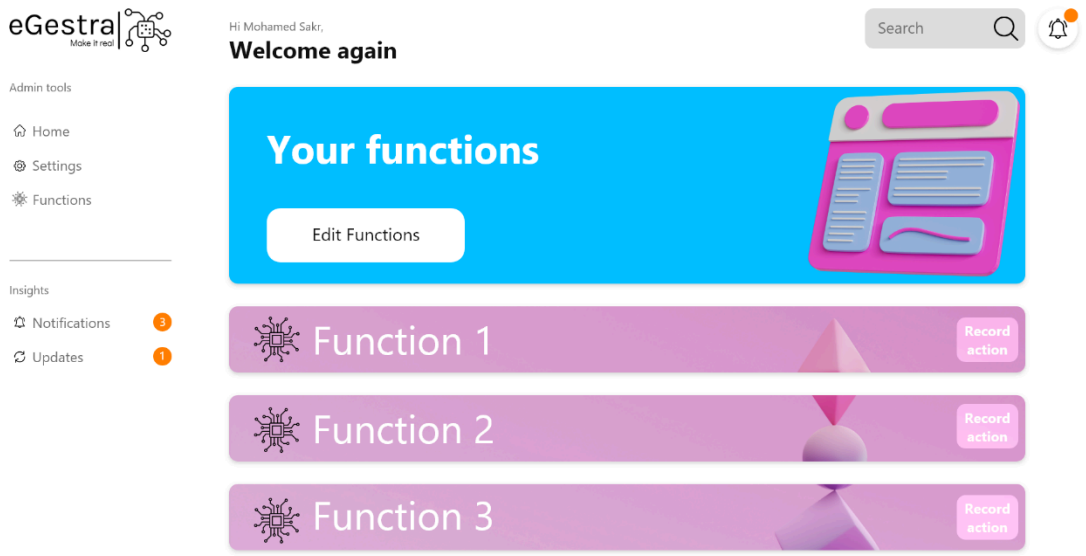

Figure 8 Home Page Sketch

Notifications Tab :

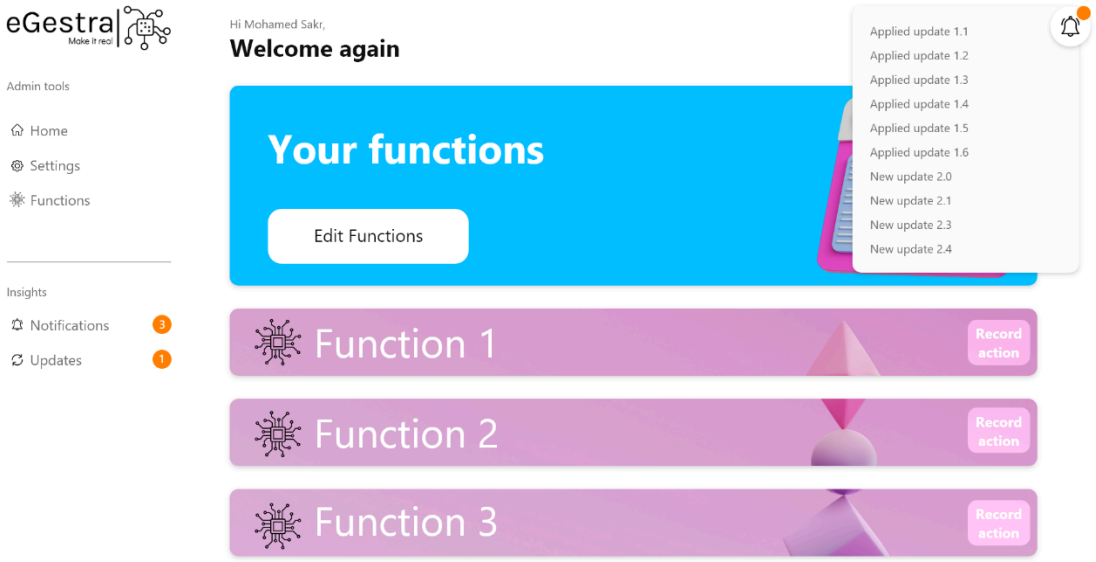

Figure 9 Notifications Tab Sketch

Functions Page:

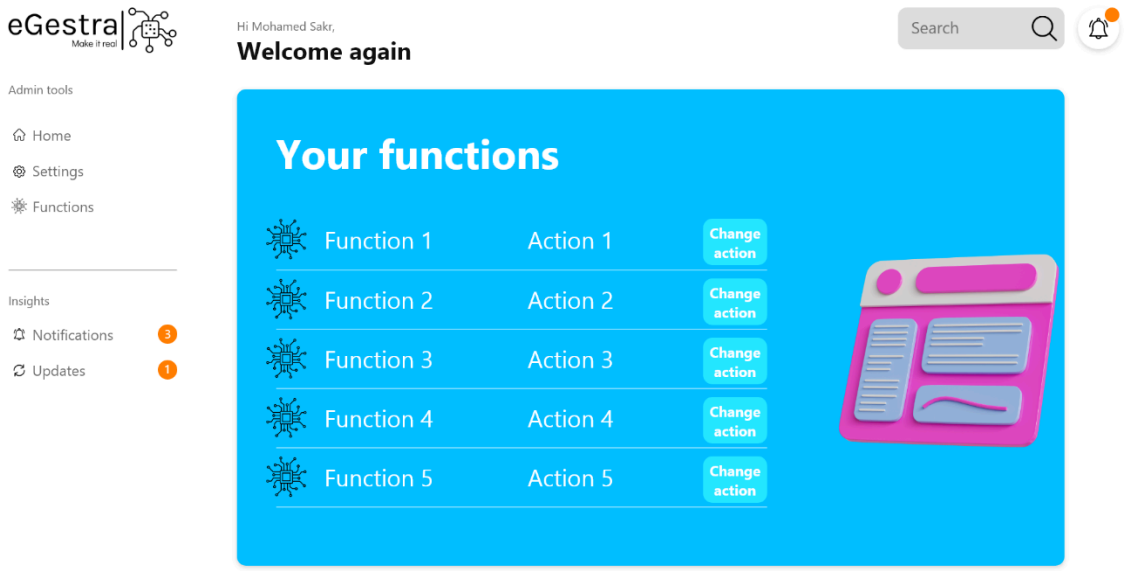

Figure 10 Functions Page Sketch

Settings Page:

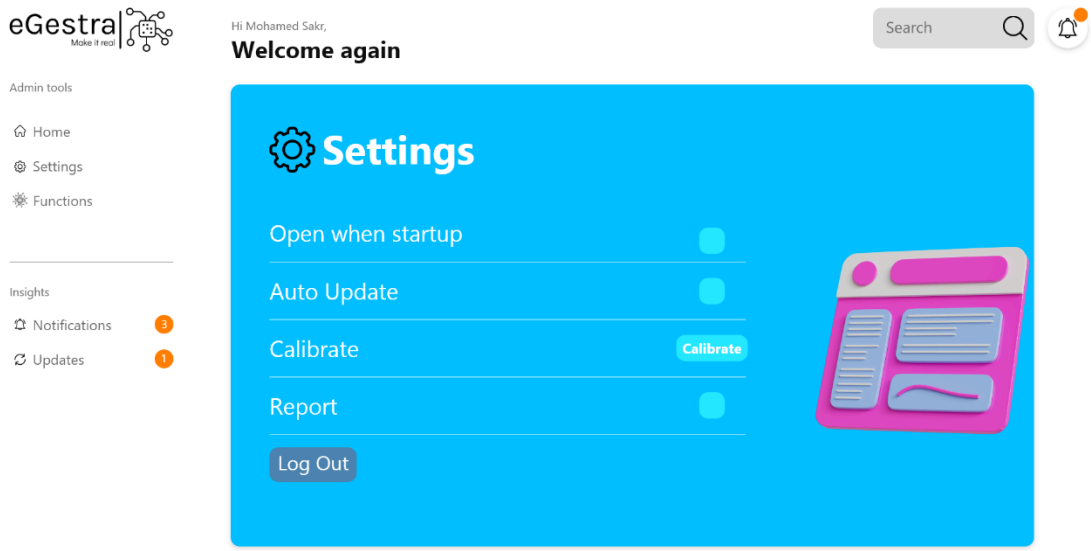

Figure 11 Settings Page Sketch

Sign Up Page:

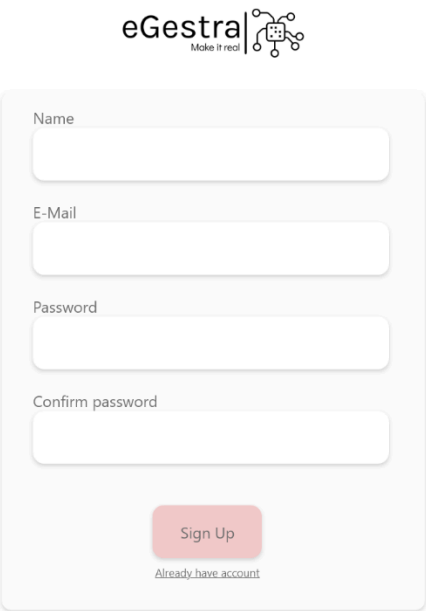

Figure 12 Sign Up Page Sketch

Sign In Page:

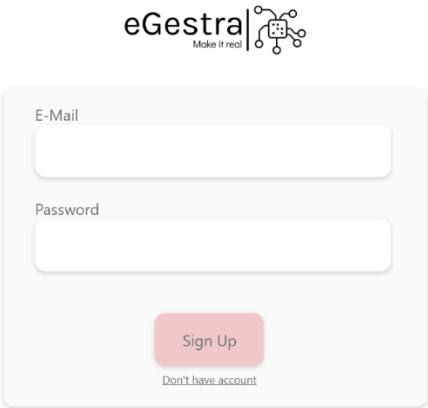

Figure 13 Sign In Page Sketch

Chapter V

5. References

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- Milvus Vector Database documentation,
- https://milvus.io/docs/
- Google Text-to-Speech API documentation.
- Related academic papers and articles on AI customer support systems and conversational agents.